Tutorial No : 4

Tutorial Title: To create and manipulate employee database by using array of structure. Data base Management using array of structure with operations Create, display, Modify, Append, Search and Sort.

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Program:

```
#include <stdio.h>
#include <string.h>

struct Employee
{
   int id;
   char name[30];
   float salary;
};

int main()
{
   struct Employee e[100];
   int n = 0, choice, i, id, found;

   while (1)
   {
      printf("\n--- EMPLOYEE DATABASE ---\n");
      printf("1. Create\n");
      printf("2. Display\n");
      printf("3. Modify\n");
      printf("4. Append\n");
      printf("5. Search\n");
      printf("6. Sort\n");
      printf("7. Exit\n");

      printf("Enter your choice: ");
      scanf("%d", &choice);
```

```
switch (choice)
{
    case 1:
        printf("Enter number of employees: ");
        scanf("%d", &n);

        for (i = 0; i < n; i++)
        {
            printf("\nEnter ID: ");
            scanf("%d", &e[i].id);

            printf("Enter Name: ");
            scanf("%s", e[i].name);

            printf("Enter Salary: ");
            scanf("%f", &e[i].salary);
        }
        break;

    case 2:
        printf("\nID\tName\tSalary\n");

        for (i = 0; i < n; i++)
        {
            printf("%d\t%s\t%.2f\n",
                  e[i].id, e[i].name, e[i].salary);
        }
        break;

    case 3:
        printf("Enter ID to modify: ");
        scanf("%d", &id);

        found = 0;

        for (i = 0; i < n; i++)
        {
            if (e[i].id == id)
            {
                printf("Enter new name: ");
```

```
            scanf("%s", e[i].name);

            printf("Enter new salary: ");
            scanf("%f", &e[i].salary);

            found = 1;
            printf("Record modified successfully.\n");
            break;
        }
    }

    if (!found)
        printf("Employee not found.\n");

    break;

case 4:
    printf("Enter ID: ");
    scanf("%d", &e[n].id);

    printf("Enter Name: ");
    scanf("%s", e[n].name);

    printf("Enter Salary: ");
    scanf("%f", &e[n].salary);

    n++;

    printf("Employee added successfully.\n");
    break;

case 5:
    printf("Enter ID to search: ");
    scanf("%d", &id);

    found = 0;

    for (i = 0; i < n; i++)
    {
        if (e[i].id == id)
```

```
        {
            printf("\nEmployee Found!\n");
            printf("ID: %d\n", e[i].id);
            printf("Name: %s\n", e[i].name);
            printf("Salary: %.2f\n", e[i].salary);

            found = 1;
            break;
        }
    }

    if (!found)
        printf("Employee not found.\n");

    break;

case 6:
    for (i = 0; i < n - 1; i++)
    {
        for (int j = i + 1; j < n; j++)
        {
            if (e[i].id > e[j].id)
            {
                struct Employee temp;

                temp = e[i];
                e[i] = e[j];
                e[j] = temp;
            }
        }
    }

    printf("Records sorted by ID.\n");
    break;

case 7:
    return 0;

default:
    printf("Invalid choice.\n");
```

```
    }
  }

  return 0;
}
```

Output :

